

Notes for ‘Differentials and applications’

Important Ideas and Useful Facts:

- (i) **Interpreting the equation linking differentials:** Consider $y = f(x)$ with derivative $y' = f'(x)$. In Leibniz notation we have

$$\frac{dy}{dx} = y' = f'(x) ,$$

which can be rewritten as an equation using differentials:

$$dy = f'(x)dx ,$$

which may be regarded as an ‘idealisation’ of the following approximation:

$$\Delta y \approx f'(x)\Delta x ,$$

where Δx is a small change in x and Δy is the corresponding small change in y . We thus get the following approximation:

$$y + \Delta y \approx y + f'(x)\Delta x .$$

This approximation improves (towards equality ‘in the limit’) as Δx approaches 0. The reason is that the tangent line to the curve is a good approximation to the curve at a given point of interest.

- (ii) **The equation of the tangent line to the natural exponential function at the y -intercept:** Euler’s number e is defined so that the slope of the tangent line to the curve $y = e^x$ at the y -intercept has slope 1. It follows that the equation of this tangent line is

$$y = x + 1 ,$$

and it becomes a good approximation to the curve near the y -intercept. A consequence is that, for small real numbers Δ , we have

$$1 + \Delta \approx e^\Delta ,$$

which is often used to simplify formulae in applications, especially when Δ represents some small periodic percentage increase.

- (iii) **The equation of the tangent line to the natural logarithm function at the x -intercept:** The slope of the tangent line to the curve $y = \ln x$ at the x -intercept has slope 1. It follows that the equation of this tangent line is

$$y = x - 1 ,$$

and it becomes a good approximation to the curve near the x -intercept. A consequence is that, for small real numbers Δ , we have

$$\ln(1 + \Delta) \approx 1 + \Delta - 1 = \Delta .$$

An application is the *Rule of Seventy*, which says that the number of years needed for the principal to double in value, when invested at compound annual interest rate $i\%$, is approximately 70 divided by i . For example, if the interest rate is 2% then one expects the principal to take about $70/2 = 35$ years to double.

Examples and derivations:

1. Use the Rule of Seventy to estimate how long it takes for the principal to double when invested at 1%, 5% and 10% compound annual interest rates respectively.

Solution: We estimate that it takes $70/1 = 70$, $70/5 = 14$ and $70/10 = 7$ years respectively for the principal to double.

2. Estimate the annual interest rate $i\%$ (to the nearest tenth of one per cent) required that a given principal should double within about 25 years.

Solution: We want i such that $25 \approx 70/i$, that is,

$$i \approx \frac{70}{25} = 2.8.$$

Thus we estimate that, using an interest rate of about 2.8%, the principal should double in about 25 years.

3. We explain where the Rule of Seventy comes from. Let P be the principal that has been invested initially at the compound interest rate of $i\%$ per annum. Let $y = y(x)$ be the value of the investment after x years, so $P = y(0)$ and

$$y = y(x) = P \left(1 + \frac{i}{100} \right)^x.$$

We want to estimate the number of years x such that $y(x) = 2P$, that is,

$$2P = P \left(1 + \frac{i}{100} \right)^x,$$

which becomes, after cancelling P from both sides,

$$2 = \left(1 + \frac{i}{100} \right)^x.$$

Taking natural logarithms of both sides, we get

$$\ln 2 = x \ln \left(1 + \frac{i}{100} \right).$$

Rearranging, and using the approximation $\ln(1 + \Delta) \approx \Delta$ noted earlier, taking $\Delta = \frac{i}{100}$, we get

$$x = \frac{\ln 2}{\ln \left(1 + \frac{i}{100} \right)} \approx \frac{\ln 2}{i/100} = \frac{100 \ln 2}{i},$$

which we expect to be a good approximation for small i . But the numerator can be approximated by 69, 70 and 72:

$$100 \ln 2 \approx 69.3 \approx 69 \approx 70 \approx 72,$$

leading to the Rule of Seventy, by dividing i into 70.

It also leads to the Rule of Sixty-nine (slightly more accurate), by dividing i into 69, and the Rule of Seventy-two (the least accurate), by dividing i into 72.

The Rule of Seventy-two is a convenient rule of thumb if the interest rate happens to exactly divide 72. For example, if the principal is invested at 6% per annum, then this rule estimates that it takes about $72/6 = 12$ years to double. It is slightly awkward to divide 6 into 69 or 70, and in both cases the answer rounds up to 12.

4. Find an equation relating the differentials dy and dx , and an associated approximation relating small changes Δy and Δx , when $y = x^2$.

Use this approximation formula to estimate the change in area of a square paddock of side length 100 metres, if the side length is extended by 10 metres, 5 metres, 1 metre and 1 cm respectively.

Compare the answers with the exact changes in area in each case.

Solution: We have $dy/dx = y' = 2x$, so that

$$dy = 2x dx \quad \text{and} \quad \Delta y \approx 2x \Delta x .$$

Let y square metres denote the area of the paddock of side length x metres, so $y = x^2$. We are considering making small changes to $x = 100$ and $y = 100^2 = 10,000$.

In the first case, $\Delta x = 10$, and, by our formula,

$$\Delta y \approx 2x \Delta x = 2 \times 100 \times 10 = 2,000 ,$$

so we expect an increase of about 2,000 square metres. But

$$y(110) - y(100) = 110^2 - 100^2 = 2,100 ,$$

so the true change in area is 2,100 square metres, which agrees with our estimate to within 100 square metres.

In the second case, $\Delta x = 5$, and, by our formula,

$$\Delta y \approx 2x \Delta x = 2 \times 100 \times 5 = 1,000 ,$$

so we expect an increase of about 1,000 square metres. But

$$y(105) - y(100) = 105^2 - 100^2 = 1,025 ,$$

so the true change in area is 1,025 square metres, which agrees with our estimate to within 25 square metres.

In the third case, $\Delta x = 1$, and, by our formula,

$$\Delta y \approx 2x \Delta x = 2 \times 100 \times 1 = 200 ,$$

so we expect an increase of about 200 square metres. But

$$y(101) - y(100) = 101^2 - 100^2 = 201 ,$$

so the true change in area is 201 square metres, which agrees with our estimate to within 1 square metre.

In the fourth case, $\Delta x = 0.01$ (since 1 cm equals 0.01 m), and, by our formula,

$$\Delta y \approx 2x \Delta x = 2 \times 100 \times 0.01 = 2 ,$$

so we expect an increase of about 2 square metres. But

$$y(100.01) - y(100) = 100.01^2 - 100^2 = 2.0001 ,$$

so the true change in area is 2.0001 square metres, which agrees with our estimate to within 0.0001 square metres.

5. Find the equation of the tangent line to the curve $y = x^2$ at the point $(3, 9)$ and use it to estimate 3.01^2 , and compare the estimate with the true value.

Solution: We have $y' = 2x$, so the slope of the tangent line is $2 \times 3 = 6$. Hence the tangent line must have equation

$$y = 6x + k ,$$

for some constant k . But the point $(3, 9)$ lies on this line, so $9 = 6(3) + k = 18 + k$, so that $k = -9$. Hence the equation of the tangent line is

$$y = 6x - 9 .$$

Using the tangent line, with input $x = 3.01$, gives the estimate

$$3.01^2 \approx 6(3.01) - 9 = 9.06 .$$

In fact, $3.01^2 = 9.0601$, which agrees with the estimate to three decimal places.

6. Find the equation of the tangent line to the curve $y = \sqrt{x}$ at the point $(9, 3)$ and use it to estimate $\sqrt{9.5}$ and $\sqrt{8.5}$, and compare with the true values.

Solution: We have $y' = \frac{1}{2\sqrt{x}}$, so the slope of the tangent line is $\frac{1}{2\sqrt{9}} = \frac{1}{6}$. Hence the tangent line must have equation

$$y = \frac{x}{6} + k ,$$

for some constant k . But the point $(9, 3)$ lies on this line, so $3 = \frac{9}{6} + k = \frac{3}{2} + k$, so that $k = \frac{3}{2}$. Hence the equation of the tangent line is

$$y = \frac{x}{6} + \frac{3}{2} = \frac{x+9}{6} .$$

Using the tangent line, with input $x = 9.5$, gives the estimate

$$\sqrt{9.5} \approx \frac{9.5+9}{6} = 3.08\dot{3} .$$

In fact, $\sqrt{9.5} \approx 3.0822$, to four decimal places, which agrees with the estimate to two decimal places.

Using the tangent line, with input $x = 8.5$, gives the estimate

$$\sqrt{8.5} \approx \frac{8.5+9}{6} = 2.91\dot{6} .$$

In fact, $\sqrt{8.5} \approx 2.9155$, to four decimal places, which again agrees with the estimate to two decimal places.

7. Find the equation of the tangent line to the curve $y = \sqrt[3]{x} = x^{1/3}$ at the point $(64, 4)$ and use it to estimate $\sqrt[3]{70}$, $\sqrt[3]{65}$ and $\sqrt[3]{63}$, and compare with the true values.

Solution: We have $y' = \frac{1}{3}x^{-2/3}$, so the slope of the tangent line is $\frac{1}{3} \times 64^{-2/3} = \frac{1}{48}$. Hence the tangent line must have equation

$$y = \frac{x}{48} + k ,$$

for some constant k . But the point $(64, 4)$ lies on this line, so $4 = \frac{64}{48} + k = \frac{4}{3} + k$, so that $k = \frac{8}{3}$. Hence the equation of the tangent line is

$$y = \frac{x}{48} + \frac{8}{3} = \frac{x + 128}{48}.$$

Using the tangent line, with input $x = 70$, gives the estimate

$$\sqrt[3]{70} \approx \frac{70 + 128}{48} = 4.125.$$

In fact, $\sqrt[3]{70} \approx 4.1213$, to four decimal places, which agrees with the estimate to almost two decimal places.

Using the tangent line, with input $x = 65$, gives the estimate

$$\sqrt[3]{65} \approx \frac{65 + 128}{48} = 4.0208\dot{3}.$$

In fact, $\sqrt[3]{65} \approx 4.0207$, to four decimal places, which agrees with the estimate to almost four decimal places.

Using the tangent line, with input $x = 63$, gives the estimate

$$\sqrt[3]{63} \approx \frac{63 + 128}{48} = 3.9791\dot{6}.$$

In fact, $\sqrt[3]{63} \approx 3.9791$, to four decimal places, which again agrees with the estimate to almost four decimal places.

8. Imagine that you are manufacturing metal ball-bearings in the shape of spheres with radius 5 mm, with up to 2% error in the radius. The problem is to use differentials to estimate the percentage error in volume.

Solution: The volume $V = V(r)$ of a sphere of radius r is given by the formula

$$V = V(r) = \frac{4\pi r^3}{3}.$$

Hence $dV/dr = 4\pi r^2$, so that $dV = 4\pi r^2 dr$, becoming the following approximation, relating small changes ΔV in the volume to small changes Δr in the radius:

$$\Delta V \approx 4\pi r^2 \Delta r.$$

In our application, $r = 5$. We are not sure whether the fluctuations in the radius cause it to be smaller or greater than $r = 5$. To capture all possibilities, we use the magnitude of Δr , and express the given information in terms of a bound on the true proportion of r that this error in the radius represents:

$$\frac{|\Delta r|}{r} \leq 0.02.$$

Working now with the proportion of V that the magnitude of ΔV represents, we have

$$\frac{|\Delta V|}{V} \approx \frac{4\pi r^2 \Delta r}{4\pi r^3/3} = \frac{3\Delta r}{r} \leq 3 \times 0.02 = 0.06.$$

Hence we estimate an error in the volume of the ball-bearings of up to about 6%.

Notice that the final answer is independent of the value of the radius r .